



NUMERICAL ANALYSIS

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Solution of System of Linear Equations

Crout's method or LU decomposition method.

It is a distinct method of solving a system of linear equations of the form $\boldsymbol{A\tilde{x} = b}$, where the matrix A is decomposed into a product of a lower triangular matrix L and an upper triangular matrix U, that is $A = LU$

Explicitly, we can write it as

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} & \dots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \dots & a_{2n} \\ \dots & \dots & \dots & \dots & \dots \\ a_{n1} & a_{n2} & a_{n3} & \dots & a_{nn} \end{bmatrix} = \begin{bmatrix} l_{11} & 0 & 0 & \dots & 0 \\ l_{21} & l_{22} & 0 & \dots & 0 \\ \dots & \dots & \dots & \dots & \dots \\ l_{n1} & l_{n2} & l_{n3} & \dots & l_{nn} \end{bmatrix} =$$

$$\begin{bmatrix} 1 & u_{12} & u_{13} & \dots & u_{1n} \\ 0 & 1 & u_{23} & \dots & u_{2n} \\ \dots & \dots & \dots & \dots & \dots \\ 0 & 0 & 0 & \dots & 1 \end{bmatrix}$$

Therefore, by LU-decomposition, the system of linear equations Ax □ b can be solved in three steps:

1- Construct the lower triangular matrix L and upper triangular matrix U.
2- Using forward substitution, solve $L\tilde{y} = b$
3- Solve $U\tilde{x} = \tilde{y}$, backward substitution.

We further elaborate the process by considering a 3×3 matrix A. We consider solving the system of equation of the form $A\tilde{x} = \tilde{b}$, where,

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}, \tilde{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} and\ b = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$$

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The matrix A is factorized as a product of two matrices L (lower triangular matrix) and U (upper triangular matrix) as follows:

$$\begin{bmatrix} l_{11} & 0 & 0 \\ l_{21} & l_{22} & 0 \\ l_{31} & l_{32} & l_{33} \end{bmatrix} \begin{bmatrix} 1 & u_{12} & u_{13} \\ 0 & 1 & u_{23} \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

L U A

$$\Rightarrow \begin{bmatrix} l_{11} & l_{11}u_{12} & l_{11}u_{13} \\ l_{21} & l_{21}u_{12} + l_{22} & l_{21}u_{13} + l_{22}u_{23} \\ l_{31} & l_{31}u_{12} + l_{32} & l_{31}u_{13} + l_{32}u_{23} + l_{33} \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

This implies

$l_{11} = a_{11},\ l_{21} = a_{21},\ \ l_{31} = a_{31};$

$l_{11}u_{12} = a_{12} \Rightarrow u_{12} = \frac{a_{12}}{l_{11}} = \frac{a_{12}}{a_{11}};$

$l_{11}u_{13} = a_{13} \Rightarrow u_{13} = \frac{a_{13}}{l_{11}} = \frac{a_{13}}{a_{11}};$

$l_{21}u_{13} + l_{22}u_{23} = a_{23} \Rightarrow u_{23} = \frac{1}{l_{22}}(a_{23} - l_{21}u_{13});$

$l_{31}u_{12} + l_{32} = a_{32} \Rightarrow l_{32} = a_{32} - l_{31}u_{12};$

$l_{31}u_{13} + l_{32}u_{32} + l_{33} = a_{33} \Rightarrow l_{33} = a_{33} - l_{31}u_{13} - l_{32}u_{23}$

Once all the value of l_{ij}'s and u_{ij}'s are obtained, we can write

$A\tilde{x} = b \ as \ LU\ \tilde{x} = b$

$Let\ U\ \tilde{x} = y, then\ L\ y = b$

$$\Rightarrow \begin{bmatrix} l_{11} & 0 & 0 \\ l_{21} & l_{22} & 0 \\ l_{31} & l_{32} & l_{33} \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} \Rightarrow \begin{bmatrix} l_{11}y_1 \\ l_{21}y_1 + l_{22}y_2 \\ l_{31}y_1 + l_{32}y_2 + l_{33}y_3 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$$

$$\Longrightarrow y_1 = \frac{b_1}{l_{11}},\ y_2 = \frac{1}{l_{22}}(b_2 - l_{21}y_1)\ \ and\ \ y_3 = \frac{1}{l_{33}}\ (b_3 - l_3 y_1 - l_{32}y_2)$$

By forward substitution we obtain, $Ux = y$

$$\Longrightarrow \begin{bmatrix} 1 & u_{12} & u_{13} \\ 0 & 1 & u_{23} \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}$$

By back substitution we get,

$x_3 = y_3$

$x_2 + u_{23}x_3 = y_2 \Longrightarrow x_2 = y_2 - u_{23}x_3$

$x_1 + u_{12}x_2 + u_{13}x_3 = y_1 \Longrightarrow x_1 = y_1 - u_{12}x_2 - u_{13}x_3$

Example 1. Solve the following system of linear equations, by Crout's method:

$$10x_1 + 3x_2 + 4x_3 = +15$$

$$2x_1 - 10x_2 + 3x_3 = 37$$

$$3x_1 + 2x_2 - 10x_3 = -10$$

Solution: In matrix form, the given system of equation can be written as

$$\begin{bmatrix} 10 & 3 & 4 \\ 2 & -10 & 3 \\ 3 & 2 & -10 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 15 \\ 37 \\ -10 \end{bmatrix}$$

Which is of form $Ax = b$. Let $A = LU$, which implies

$$\begin{bmatrix} 10 & 3 & 4 \\ 2 & -10 & 3 \\ 3 & 2 & -10 \end{bmatrix} = \begin{bmatrix} l_{11} & 0 & 0 \\ l_{21} & l_{22} & 0 \\ l_{31} & l_{32} & l_{33} \end{bmatrix} \begin{bmatrix} 1 & u_{12} & u_{13} \\ 0 & 1 & u_{23} \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} l_{11} & l_{11}u_{12} & l_{11}u_{13} \\ l_{21} & l_{21}u_{12} + l_{22} & l_{21}u_{13} + l_{22}u_{23} \\ l_{31} & l_{31}u_{12} + l_{32} & l_{31}u_{13} + l_{32}u_{23} + l_{33} \end{bmatrix}$$

$$\Longrightarrow l_{11} = 10, l_{21} = 2, l_{31} = 3; u_{12} = \frac{3}{10}, u_{13} = \frac{4}{10};$$

$$l_{21}u_{12} + l_{22} = -10 \Longrightarrow l_{22} = -10 - 2 \times \frac{3}{10} = \frac{106}{10};$$

$$l_{21}u_{13} + l_{22}u_{23} = 3 \Longrightarrow u_{23} = \frac{3 - 2 \times \frac{4}{10}}{-\frac{106}{10}} = -\frac{11}{53};$$

$$l_{31}u_{12} + l_{32} = 2 \Longrightarrow l_{32} = 2 - l_{31}u_{12} = 2 - 3 \times \frac{3}{10} = \frac{11}{10};$$

$$l_{31}u_{13} + l_{32}u_{23} + l_{33} = -10 \Longrightarrow l_{33} = -10 - l_{31}u_{13} - l_{32}u_{23}$$

$$= -10 - 3 \times \frac{4}{10} + \frac{11}{10} \times \frac{11}{53} = -\frac{1163}{106}$$

Therefore, we get,

$$L = \begin{bmatrix} 10 & 0 & 0 \\ 2 & \frac{-106}{10} & 0 \\ 3 & \frac{11}{10} & \frac{-1163}{106} \end{bmatrix} and\ U = \begin{bmatrix} 1 & \frac{3}{10} & \frac{4}{10} \\ 0 & 1 & \frac{-11}{53} \\ 0 & 0 & 1 \end{bmatrix}$$

Now, let $Ux = y$, then Ly=b implies

$$\begin{bmatrix} 10 & 0 & 0 \\ 2 & \frac{-106}{10} & 0 \\ 3 & \frac{11}{10} & \frac{-1163}{106} \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} 15 \\ 37 \\ -10 \end{bmatrix}$$

$$10y_1 = 15 \Longrightarrow y_1 = \frac{3}{2}$$

This implies $2y_1 - \frac{106}{10} y_2 = 37 \Longrightarrow y_2 = \frac{-170}{53}$

$$y_1 + \frac{11}{10} y_2 - \frac{1163}{106} y_3 = -10 \Longrightarrow y_3 = 1$$

Thus, $y = \begin{bmatrix} 3/2 \\ \frac{-170}{53} \\ 1 \end{bmatrix}$ and $Ux = y$ gives

$$\begin{bmatrix} 1 & \frac{3}{10} & \frac{4}{10} \\ 0 & 0 & \frac{-11}{53} \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} \frac{3}{2} \\ \frac{-170}{53} \\ 1 \end{bmatrix}$$, which implies

$$x_1 + \frac{3}{10} x_2 - \frac{4}{10} x_3 = \frac{3}{2} \quad ; \quad x_2 - \frac{11}{53} x_3 = \frac{-170}{53} \quad ; \quad x_3 = 1$$

By back substitution, we get,

$x_3 = 1 \qquad ; x_2 = \frac{11\times1}{53} - \frac{170}{53} = -1 \ ;$

$x_1 = \frac{3}{2} - \frac{3}{10}x_2 - \frac{4}{10}x_3 = \frac{3}{2} - \frac{3}{10} \times (-3) - \frac{4}{10} \times 1 = 2;$

Therefore, the required solution by Crout's method (LU decomposition method) is $x_1 = 2,\ x_2 = -3\ , x_3 = 1$

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Example 2. Solve the following system of linear equations by Crout's Method (LU factorization or decomposition method):

$9x_1 + 3x_2 + 3x_3 + 3x_4 = 24$

$3x_1 + 10x_2 - 2x_3 - 2x_4 = 17$

$3x_1 - 2x_2 + 18x_3 + 10x_4 = 45$

$3x_1 - 2x_2 + 10x_3 + 10x_4 = 29$

Solution: The given system of equation can be written in matrix form as

$$\begin{bmatrix} 9 & 3 & 3 & 3 \\ 3 & 10 & -2 & -2 \\ 3 & -2 & 18 & 10 \\ 3 & -2 & 10 & 10 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 24 \\ 17 \\ 45 \\ 29 \end{bmatrix}$$

Let $\begin{bmatrix} 9 & 3 & 3 & 3 \\ 3 & 10 & -2 & -2 \\ 3 & -2 & 18 & 10 \\ 3 & -2 & 10 & 10 \end{bmatrix} = \begin{bmatrix} l_{11} & 0 & 0 & 0 \\ l_{21} & l_{22} & 0 & 0 \\ l_{31} & l_{32} & l_{33} & 0 \\ l_{41} & l_{42} & l_{43} & l_{44} \end{bmatrix} \begin{bmatrix} 1 & u_{12} & u_{13} & u_{14} \\ 0 & 1 & u_{23} & u_{24} \\ 0 & 0 & 1 & u_{34} \\ 0 & 0 & 0 & 1 \end{bmatrix}$

Comparing, we get,

$l_{11} = 9, l_{21} = 3, l_{31} = 3, l_{41} = 3;$

$l_{11}u_{21} = 3 \Longrightarrow u_{12} = \frac{1}{3}$.Similarly, $u_{13} = u_{14} = \frac{1}{3}$;

$l_{21}u_{12} + l_{22} = 10 \Longrightarrow l_{22} = 10 - l_{21}u_{12} = 10 - 3 \times \frac{1}{3} = 9$

$l_{21}u_{13} + l_{22}u_{23} = -2 \Longrightarrow u_{23} = \frac{-2-l_{21}u_{13}}{l_{22}} = -\frac{1}{3}$

$l_{21}u_{14} + l_{22}u_{24} = -2 \Longrightarrow u_{24} = \frac{-2-l_{21}u_{14}}{l_{22}} = -\frac{1}{3}$

$l_{31}u_{12} + l_{32} = -2 \Longrightarrow l_{32} = -2 - l_{31}u_{12} = -3$

$l_{31}u_{13} + l_{32}u_{23} + l_{33} = 18 \Longrightarrow l_{33} = 18 - l_{31}u_{13} - l_{32}u_{23} = 16$

$l_{31}u_{14} + l_{32}u_{24} + l_{33}u_{34} = 10 \Longrightarrow u_{34} = \frac{10-l_{31}u_{14}-l_{32}u_{24}}{l_{33}} = \frac{1}{2}$

$l_{41}u_{12} + l_{42} = -2 \Longrightarrow l_{42} = -2 - l_{41}u_{12} = -3$

$l_{41}u_{13} + l_{42}u_{23} + l_{43} = 10 \Longrightarrow l_{43} = 10 - l_{41}u_{13} - l_{42}u_{23} = 8$

$l_{41}u_{14} + l_{42}u_{24} + l_{42}u_{34} + l_{44} = 10$

$\Longrightarrow l_{44} = 10 - l_{41}u_{14} - l_{42} - l_{42}u_{24} - l_{43}u_{34} = 4$

Therefore, we get

$$L = \begin{bmatrix} 9 & 0 & 0 & 0 \\ 3 & 9 & 0 & 0 \\ 3 & -3 & 16 & 0 \\ 3 & -3 & 8 & 4 \end{bmatrix} \text{ and } U = \begin{bmatrix} 1 & 1/3 & 1/3 & 1/3 \\ 0 & 0 & -1/3 & -1/3 \\ 0 & 0 & 1 & 1/2 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Forward substitution gives

$9y_1 = 24 \Longrightarrow y_1 = \frac{8}{3}$

$3y_1 + 9y_2 = 17 \Longrightarrow y_2 = 1$

$3y_1 - 3y_2 + 16y_3 = 45 \Longrightarrow y_3 = \frac{5}{2}$

$3_{y1} - 3y_2 + 8y_3 + 4y_4 = 29 \Longrightarrow y_4$

Thus, $y = \begin{bmatrix} 8/3 \\ 1 \\ 5/2 \\ 1 \end{bmatrix}$ and $Ux = y$ gives

$$\begin{bmatrix} 1 & 1/3 & 1/3 & 1/3 \\ 0 & 1 & -1/3 & -1/3 \\ 0 & 0 & 1 & 1/2 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ y_3 \end{bmatrix} = \begin{bmatrix} 8/3 \\ 1 \\ 5/2 \\ 1 \end{bmatrix}$$

By back substitution we get,

$x_4 = 1$; ; $x_3 + \frac{1}{2}x_4 = \frac{5}{2} \Longrightarrow x_3 = 2$

$x_2 - \frac{1}{3}x_3 - \frac{1}{3}x_4 = 1 \Longrightarrow x_2 = 2$; $x_1 + \frac{1}{3}x_2 + \frac{1}{3}x_3 + \frac{1}{3}x_4 = \frac{8}{3} \Longrightarrow x_1 = 1$

Therefore, the required solution by Crout's method

is $x_1 = 1, x_2 = 2, x_3 = 2, x_4 = 1$

H.W/ Solve the following system of linear equations by Crout's Method (LU factorization or decomposition method):

1. $$\begin{aligned} 2x1+3x2+x3&=1\\ -2x1+2x2+4x3&=1\\ x1+3x2+8x3&=-6 \end{aligned}$$

2. $$\begin{aligned} 2x1+x2+x3&=4\\ x1+2x2+x3&=2\\ x1+x2+2x3&=-9 \end{aligned}$$

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